

PROGRAM:

1. ARITHMETIC OPERATORS

```
print("==== 1. ARITHMETIC OPERATORS ====")
```

```
a = 10
```

```
b = 3
```

```
print("Addition    :", a + b) # +
```

```
print("Subtraction :", a - b) # -
```

```
print("Multiplication :", a * b) # *
```

```
print("Division     :", a / b) # /
```

```
print("Floor Division :", a // b) # //
```

```
print("Modulus      :", a % b) # %
```

```
print("Exponent     :", a ** b) # **
```

2. ASSIGNMENT OPERATORS

```
print("\n==== 2. ASSIGNMENT OPERATORS ====")
```

```
x = 10
```

```
print("x =", x)
```

```
x += 5
```

```
print("x += 5 :", x)
```

```
x -= 3
```

```
print("x -= 3 :", x)
```

```
x *= 2
```

```
print("x *= 2 :", x)
```

```
x /= 4
```

```
print("x /= 4 :", x)
```

```
x //= 2
```

```
print("x //= 2 :", x)
```

```
x %= 3
```

```
print("x %= 3 :", x)
```

```
x **= 2
```

```
print("x **= 2 :", x)
```

3. COMPARISON OPERATORS

```
print("\n===== 3. COMPARISON OPERATORS =====")
```

```
p = 10
```

```
q = 20
```

```
print("p == q :", p == q) # Equal
```

```
print("p != q :", p != q) # Not equal
```

```
print("p > q :", p > q) # Greater than
```

```
print("p < q :", p < q) # Less than
```

```
print("p >= q :", p >= q) # Greater than or equal
```

```
print("p <= q :", p <= q) # Less than or equal
```

4. LOGICAL OPERATORS

```
print("\n===== 4. LOGICAL OPERATORS =====")
```

```
age = 20
```

```
has_id = True
```

```
print("AND :", age >= 18 and has_id)
```

```
print("OR :", age >= 18 or has_id)
```

```
print("NOT :", not has_id)
```

5. IDENTITY OPERATORS

```
print("\n===== 5. IDENTITY OPERATORS =====")
```

```
list1 = [1, 2, 3]
```

```
list2 = list1
```

```
list3 = [1, 2, 3]
```

```
print("list1 is list2  :", list1 is list2)
```

```
print("list1 is list3  :", list1 is list3)
```

```
print("list1 is not list3 :", list1 is not list3)
```

6. MEMBERSHIP OPERATORS

```
print("\n===== 6. MEMBERSHIP OPERATORS =====")
```

```
numbers = [10, 20, 30, 40, 50]
```

```
print("20 in numbers  :", 20 in numbers)
```

```
print("60 in numbers  :", 60 in numbers)
```

```
print("60 not in numbers :", 60 not in numbers)
```

OUTPUT:

===== 1. ARITHMETIC OPERATORS =====

Addition : 13

Subtraction : 7

Multiplication : 30

Division : 3.3333333333333335

Floor Division : 3

Modulus : 1

Exponent : 1000

===== 2. ASSIGNMENT OPERATORS =====

x = 10

x += 5 : 15

x -= 3 : 12

x *= 2 : 24

x /= 4 : 6.0

x //= 2 : 3.0

x %= 3 : 0.0

x **= 2 : 0.0

===== 3. COMPARISON OPERATORS =====

p == q : False

p != q : True

p > q : False

p < q : True

p >= q : False

p <= q : True

===== 4. LOGICAL OPERATORS =====

AND : True

OR : True

NOT : False

===== 5. IDENTITY OPERATORS =====

list1 is list2 : True

list1 is list3 : False

list1 is not list3 : True

===== 6. MEMBERSHIP OPERATORS =====

20 in numbers : True

60 in numbers : False

60 not in numbers : True